AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# MST224
# Mathematical Methods

## Final Examination
## Fall Semester 2025/2026

## Date: 12/Jan./2026

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| **Number of Exam Pages:** 3 (including this cover sheet) | **Time Allowed:** 3H |

## Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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## PART 1

**1. [5×2 marks] Choose the correct answer:**

**1.1** If $f(x,y) = x^3y^2 - 2x^2y + 3x$, then $f_x(2,-1) =$

a) 23
b) $-23$
c) 7
d) $-7$
e) None of the above.

**1.2** The solution of the boundary value problem $y'' + 4y = 0; y(0) = 4, y\left(\frac{\pi}{2}\right) = 1$, is:

a) $y = -\cos 2x - 4\sin 2x$
b) $y = \cos 2x + 4\sin 2x$
c) $y = -4\cos 2x - \sin 2x$
d) $y = 4\cos 2x + \sin 2x$
e) None of the above.

**1.3**: Let the position vector of a point $P$ is given by $\mathbf{r}(t) = 2t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k}$. The acceleration $\mathbf{a}(t)$ at $t = 1$ is:

a) $6\mathbf{j} + 6\mathbf{k}$.
b) $2\mathbf{i} + 6\mathbf{j} + 3\mathbf{k}$.
c) $2\mathbf{i} + 3\mathbf{j} + \mathbf{k}$.
d) $6\mathbf{i} + 6\mathbf{k}$.
e) None of the above.

**1.4** Let $V(x, y, z) = xy^2z^3$, then $\nabla \times \nabla V =$

a) $y^2z^3\mathbf{i} + 2xyz^3\mathbf{j} + 3xy^2z^2\mathbf{k}$.
b) 0.
c) $2xz^3\mathbf{j} + 6xy^2z\mathbf{k}$.
d) $\mathbf{i} + 2xz^3\mathbf{j} + 6xy^2z\mathbf{k}$.
e) None of the above.

**1.5** The length of a quarter-circle path of radius $R = 4$ is:

a) $2\pi$.
b) $\frac{\pi}{2}$.
c) $4\pi$.
d) $\frac{\pi}{4}$.
e) None of the above.

**2. [5×2 marks] Determine whether following statements are True or False and <u>justify</u> your answer in both cases.**

**2.1** If $f(x,y) = y^2 + x^3$, then $\nabla \times \nabla f = \mathbf{0}$.

**2.2** The partial differential equation $U_{tt} = 3U_{xx}$ describes the conduction of heat, where $U$ is the temperature and 3 is the thermal diffusivity.

**2.3** The vector field $\mathbf{F} = -2x\,\mathbf{i} + 6y\,\mathbf{j} - 8z\,\mathbf{k}$ is a gradient field with an associated scalar potential field $U = x^2 - 3y^2 + 4z^2$.

**2.4** The vector $\mathbf{b} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ is a unit vector.

**2.5** For a non-zero vector $\mathbf{a}$, then $\mathbf{a} \times \mathbf{a} = \mathbf{0}$

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## PART 2

**Q1: [5 marks]** Find and classify the critical points of

$$f(x,y) = x^2 + 2y^2 - 4x + 4y + 6.$$

**Q2: [5 marks]** Given the function

$$f(x,y) = 3 - \frac{x^2}{10} + \frac{xy^2}{10}$$

i. Compute $\nabla f(3,-1)$.

ii. Find the slope of the function $f(x,y)$ at $(3,-1)$ in the direction of the (non-unit) vector $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$.

**Q3: [5 marks]** Consider the following differential equation,

$$\frac{dy}{dx} = \frac{1}{3y^2(x-1)}, \qquad x > 1.$$

i. Find the general solution.

ii. Given that the solution is the curve $y$ passes though the point $P(2,-1)$ and the point $Q(q,1)$, find the exact value of $q$.

## PART 3

**Q1: [5 marks]** Use the chain rule to find $\partial w/\partial p$ and $\partial w/\partial q$, if

$$w = r^3 + s^3, \quad \text{with} \quad r = pq^2, s = p^2 \sin q.$$

**Q2: [4 marks]** Find the area integral of the function $f(x,y) = y^3 + 4x$, over the region enclosed by the graphs $x = y^2$ and $x = 2y$.

**Q3: [6 marks]** Consider the non-linear system of differential equations

$$\dot{x} = 10x - 5xy,$$
$$\dot{y} = 3y + xy - 3y^2.$$

(a) Find the equilibrium points of the system.

(b) Find the Jacobian matrix of the system.

(c) In the neighborhood of each equilibrium point:

- find the linear system of differential equations that gives the approximate behavior of the system near the equilibrium points.
- find the eigenvalues of the Jacobian matrix.
- use the eigenvalues to classify the equilibrium point of the linearized system.

**End of questions**



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